Whisper Speech-to-Text Implementation – Brief Report

1. Prompted vs. Unprompted Transcription

The audio was first transcribed without a prompt to establish a baseline and was then transcribed again using a contextual prompt containing terms related to the Moon landing. Both approaches produced readable transcriptions. The prompted approach gave Whisper additional context that could help it recognise specialised terminology and proper names when they occur in the audio. In this sample, the prompt did not cause unrelated contextual terms to be inserted into the transcription, showing that the prompt mainly acted as guidance. For business meetings, contextual prompts could therefore be useful when specific company names, technical terms, acronyms, or people's names are expected.

2. Benefits of Chunking Long Audio

Audio chunking makes long recordings easier to process by dividing them into smaller, manageable sections. In this project, the 86.2-second sample was divided into three chunks of approximately 30 seconds each to demonstrate the technique. For longer meetings, larger intervals such as 10-minute chunks could be used. Chunking also makes it easier to identify and retry a failed section without repeating the entire transcription. To preserve the original timeline, each chunk's starting offset was added to the timestamps returned by Whisper.

3. Challenges Faced

One challenge was ensuring that the audio file and required tools were correctly configured before transcription. The audio had to be accessible and readable through pydub, which relies on FFmpeg for audio processing. Another challenge was handling timestamps after splitting the audio. Without adjustment, each chunk would restart at 00:00, making the combined transcript inaccurate. This was solved by calculating the offset of each chunk and adding it to the segment timestamps. Separate formatting was also required for human-readable TXT timestamps and SRT subtitle timestamps.

4. Recommendations for Improving Accuracy

Accuracy can be improved by providing relevant contextual prompts containing verified names, technical terms, abbreviations, and vocabulary expected in the recording. Good-quality audio with clear speech and minimal background noise should also improve results. For long recordings, chunk sizes should retain enough conversational context while keeping each segment manageable. Important transcripts should be reviewed for names, numbers, technical terminology, and unclear speech. In a production solution, additional validation and retry logic could automatically identify and reprocess failed chunks.

Conclusion

The lab successfully demonstrated an end-to-end speech-to-text workflow using OpenAI Whisper: validating audio, generating basic and context-guided transcriptions, comparing the results, splitting audio into chunks, preserving timestamps, and exporting the final transcript as TXT, SRT, and JSON. The same approach can be extended to longer business meetings and other applications such as subtitles, meeting notes, accessibility, and searchable audio archives.